Course Map

For teenagers and one adult

Ten hands-on lessons that build to a Tesla coil

This is a course with a machine at the end of it. Every lesson exists because the Tesla coil in Lesson 11 will not work unless you understand the thing the lesson teaches — and each one is built, measured and broken rather than read about. The order is not negotiable: a coil is a pair of coupled resonant circuits fed by a spark gap, and each of those words is a lesson.

The through-line

#	Lesson	The idea	Why the coil needs it
1	Charge	Charge exists, comes in two signs, and is conserved. Build an electroscope.	The whole machine moves charge on and off a top electrode.
2	Fields	A field is how a force reaches across empty space. Map one.	The spark, the corona and the capacitor are all field effects.
3	Ohm's law	Voltage, current and resistance, measured on a real circuit.	You cannot size a resistor or read a meter without it.
4	Electromagnets	A current makes a magnetic field. Wind one and lift things.	Every coil in the machine is this.
5	Induction	A *changing* magnetic field makes a voltage.	This is the entire secret of the coil.
6	Transformers	Two coils sharing a field trade voltage for current.	The neon-sign transformer, and the coil itself.
7	Capacitors	Storing energy in an electric field, and the RC time constant.	The tank capacitor stores the shot the coil fires.
8	Resonance	An LC circuit has one frequency it likes. Measure it.	Both halves of the coil are tuned to the same note.
9	Coupling	Two tuned circuits pass energy back and forth.	Why a Tesla coil is not just a transformer.
10	The spark gap	Air breaks down at a voltage and becomes a switch.	It is the fastest switch in the machine, and the loudest.
11	**Build the coil**	Assemble, tune and fire it.	—

How each lesson runs

Predict first Every lesson opens with a question and a box to commit an answer in. A prediction that turns out wrong teaches more than a demonstration that was never in doubt — but only if it was written down first.

Build it They do the building. An adult who assembles the apparatus and then demonstrates it has taught nothing except that physics is something other people do.

Measure it Every lesson produces a number. A meter reading is the difference between a science lesson and a magic trick.

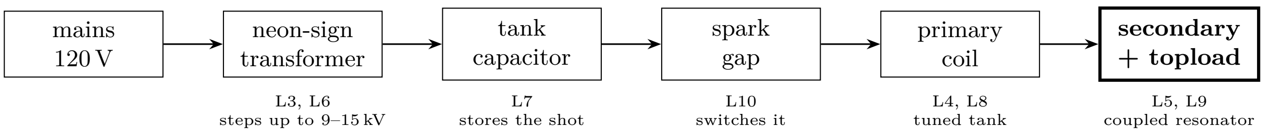
Break it Change one thing and predict the change. Double the turns. Halve the capacitor. Reverse the magnet.

Then the maths The equation comes after the measurement, as the compact way to say what they already saw. Never before.

Who does what

Lessons 1–9	Battery and low-voltage only. Nothing in these lessons can hurt anybody. The boys build, wire, measure and record; the adult asks questions.
Lesson 10 onward	Mains-derived high voltage enters the course. **The adult owns the plug and every energizing decision, permanently.** The boys still do the mechanical build — winding the secondary, cutting pipe, assembling the capacitor bank, laying out the base — because that is most of the work and all of it is safe with the thing unplugged.
Non-negotiable	Nobody is ever alone with the machine energized or energizable. The **Safety** sheet is read start to finish before Lesson 10, and it is posted on the wall where the coil is built.

The machine you are building



The one sentence version

A transformer charges a capacitor; a spark gap dumps that capacitor into a tuned coil; that coil rings at its resonant frequency and hands its energy, over several cycles, to a second coil tuned to the same frequency — which is why the output voltage is far higher than the transformer alone could ever produce. Everything in Lessons 1–10 is one word of that sentence.